

Townsend's 2026 Monthly Dust Loss Model

Purpose

This model is used to estimate the percentage of potential monthly photovoltaic (PV) energy sacrificed due to dust soiling.

PV financing decisions are based on long-term energy estimates, and popular PV modeling software such as PVsyst include monthly inputs for soiling losses – snow in the winter, dust in the summer. This modeling tool relies principally on monthly rainfall to determine a changing pattern of soiling throughout the year. That pattern is also influenced by five other factors:

1. The daily soiling ramp rate (seasonally adjustable).
2. The number of manual array washings, up to two per year.
3. The influence of snow, where any month with at least one 1" snow event will reset the dust loss to the minimum of 1% or whatever lesser amount the model may have otherwise calculated for that month. For two snow events, the dust loss is automatically reset to zero.¹
4. For bifacial arrays, the back side's contribution is presumed to have no appreciable dust buildup. The full array's monthly dust loss is reduced a little by the percentage of energy contributed from the always-clean back side.
5. For locations susceptible to seasonal pollens, mold, and similar biological sticky particulates, the model adds a +2% loss from March through October. A manual wash on the 1st day of any of these months will initially zero out the loss, but since it will ramp right up with no grace period, the pollen/mold loss for a wash month is treated as an additional 1% loss, not a 2% loss. Without any subsequent manual washings, any subsequent pollen-affected months are treated as a 2% additional dust loss.

Rules

The following table lists the core assumptions applied to each month, subject to the superseding influence of snow, which sets the dust loss to no more than 1% after one snow

¹ Townsend's snow loss measurements during 2013-15 for seven fixed-tilt orientations at Golden, CO showed 6 of the test period's 24 months had one snow event and 32 months had two events. For the single-event months, the average monthly snow loss was, rounded to the nearest integer percentage, 5%, and for 2-event months, the average loss was 21%. For trackers, the loss would be about 1/3rd of the amount experienced by a fixed-tilt structure, so tying the dust cleaning potential of snow is more closely related to the number of events than to the widely varying fixed-tilt vs tracker energy loss associated with those events. One snow event of at least 1" is viewed as sufficient to almost all of the dust off; here, we assume the residual dust won't exceed 1% after one event. Given the large jump in energy loss associated with a second event, we further assume a second snow event in the same month will be sufficient to zero out any dust loss.

event and 0% for two or more snow events. For all months, the total loss = snow + dust – (snow x dust), where, again, some winter months may see their dust lost set to 0% or 1%, or somewhere in between, depending on the number of snow events and the otherwise applicable dust loss calculation for the affected snowy months.

Table 1. Precipitation-based soiling rules

Monthly precipitation, inches (mm)	Soiling
≥4 (102)	0%
≥2 and <4 (51-102)	1%
≥0.5 and <2 (13-51)	If prior month precip. ≥2", add 8 days x seasonal ramp rate to prior month's soiling. If prior month precip. <2", add 15 days x seasonal ramp rate to prior month's soiling.
<0.5 (13)	Add product of number of days per month x seasonal ramp rate applicable to current month to prior month's soiling.

A few other non-precipitation-related rules are applied, too.

1. As with the Kimber et al. model, a maximum monthly loss of 30% is assumed.ⁱ
2. A starting month must be chosen. We need a consistent, wrapped pattern of long-term average monthly dust losses applicable to all years. To do this, the month with the most precipitation is chosen as the starting month. The soiling applicable to that month is subject to the same rules described above, with one exception: if the wettest, starting month's precipitation is less than 2 inches, the soiling for that month is set to 2%, consistent with the NREL ("NLR") default year-round assumption in their PVWATTS program.²
3. A manual wash is assumed to occur on the first day of a month. Unlike the Kimber et al. modelⁱ, this model does not include a "grace period" of 0% soiling after a manual wash. The dust loss in a manual wash month will still be calculated based on a half month of accumulation, at the daily ramp rate applicable to that season.

If one or two manual washings per year are selected, the model does an exhaustive search, testing each month in sequence as the date of the first potential wash, then recomputing the *annual* energy loss associated with each potential choice. After checking each month, the wash month with the lowest annual soiling loss is deemed to be optimal. The same logic is used to determine when best to do a second wash. It is possible that the second-best time to do a manual wash may occur chronologically in a month earlier than the best first wash. **The incremental annual loss reduction value of the "second" wash is predicated on committing to also performing the more impactful "first" wash in its prescribed month.** Here, the labeling of

² <https://pvwatts.nrel.gov/> See "Loss Calculator" in the program's "System Info" tab.

“best” and “second-best” is probably more accurate than saying “first” or “second.” If wash #1 is prescribed for August and wash #2 is prescribed for July, the low annual soiling loss being sought for two manual washes will only be attained if both the July and August washings occur.

Inputs

1. Twelve monthly precipitation amounts, specified in units of inches or millimeters.
2. Four seasonal daily dust accumulation (ramp) rates. Four rate choices are available per season, and the model has been automated to offer recommendations based on a combination of the annual rainfall and a qualitative determination of four local adder effects that increase soiling. The four rate choices bracket a wide and realistic range of observed daily dust accumulation rates:
 - a. 0.025%: mostly applies to ultra-sandy, inorganic particulates, <10"/yr rain
 - b. 0.05%: scrub desert, 10-15"/yr rain
 - c. 0.1%: typical default, safely applicable in many climates with >15"/yr rain
 - d. 0.15%, typical of summer/fall agricultural dust
3. Soiling adders. The above rates may be ratcheted up if certain conditions exist. Five qualitative choices are available; four of these can each add a +1 increment to the otherwise applicable rate in certain seasons, subject to the maximum accumulation rate of 0.15%/day. The fifth adder is different, as described below. The five possible adders include:
 - a. Agricultural, especially irrigated.
 - b. Under a commercial airport flight path and within 5 km of a runway. For guidance, a flight path is assumed to be 2 km wide, so this condition would apply to any 10 km² rectangle on either end of a main runway. Most airports seem to have 4 to 6 such flight paths for such combustion fallout.
 - c. Prone to heavy bird droppings. Practically, this implies the array is close to oceans, lakes, and rivers where large flocks are known to congregate.
 - d. Congested urban; prone to automotive and industrial emissions. For simplicity, this condition is advised within cities of at least 250,000 people.
 - e. Pollen/mold. This adder can be checked if a site is affected by heavy pollen or stubborn surface contaminants such as musts or lichens, as have been observed in California's wine country or in humid southeastern states. If this adder is checked, from March through October, an automatic +2% loss will be added on top of whatever other dust losses the model has calculated. If a manual wash occurs during a pollen-affected month, the 2% loss is zeroed out, but only on the first day of the wash month. The pollen loss during a wash month is therefore set to a +1% additional loss.

4. Number of manual washes, from 0 to 2 per year. The model finds the most effective months to perform the wash(es). In some cases, the second-best wash month may occur chronologically before the optimal wash month, but the reduced losses associated with the second-best wash are only achievable if the best wash is also done in its later month.
5. Bifacial: If bifacial, a separate calculation linked to the snow model is used to determine the monthly percentage of generation attributable to the essentially unsoiled back side, with the overall dust loss reduced accordingly.

Ramp Rates

The following table shows the recommended seasonal daily ramp rates as a function of annual rainfall in the absence of any local soiling adder effects. These preliminary rates are then adjusted upward by the number of adders applicable to a given site.

Table 2. Seasonal ramp rates for various annual precipitation levels

Annual rainfall	Wettest quarter	2 nd wettest qtr.	2 nd driest qtr.	Driest quarter
<10"	0.025%	0.025%	0.025%	0.025%
10-15"	0.025%	0.025%	0.025%	0.05%
15-25"	0.025%	0.05%	0.1%	0.15%
25-40"	0.05%	0.05%	0.1%	0.1%
>40"	0.1%	0.1%	0.1%	0.1%

The following table amends the above rates by the number of rate increments based on the number of adders that apply. The final rate is clamped at a maximum of 0.15%/day, regardless of how many adders apply. As noted earlier, the fifth soiling adder is not a daily rate term, but rather, is an additional fixed 1% or 2% adder during March-October, depending on the number of manual washes selected.

Table 3. Ramp rate increments for various adder conditions

Quarter	0 adders	1 adder	2 adders	3 adders	4 adders
Wettest	No change	No change	No change	+1	+2
2 nd wettest	No change	No change	+1	+2	+2
2 nd driest	No change	+1	+1	+2	+3
Driest	No change	+1	+2	+3	+3

As an example, consider a location such as Oakland, California, with an average annual rainfall of 24 inches. The urban adder would apply based on the city's population of 440,000 since this exceeds the nominal 250,000 population threshold. Depending on where in the city a project was being considered, the bird presence and airport flight path adders could plausibly apply, but for this example, we'll assume only one soiling adder will apply. Per the second table above, the

“1 adder” column adds one rate increment to the two driest quarters. This increases the fall rate from 0.1%/day to 0.15%/day. The summer rate was already at its maximum allowable value, so it remains unchanged.

Table 4. Oakland, California seasonal rate recommendation with one soiling adder

Oakland rainfall	Wettest (winter, 13.7")	2 nd wettest (spring, 5.6")	2 nd driest (fall, 4.5")	Driest (summer, 0.2")
15-25"	0.025%	0.05%	0.15%	0.15%

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ⁱ Kimber, A., Mitchell, L., Nogradi, S. and Wenger, H. (2006) The Effect of Soiling on Large Grid-Connected Photovoltaic Systems in California and the Southwest Region of the United States. Proceedings of the 4th IEEE World Conference on Photovoltaic Energy Conference, Waikoloa, 2391-2395.
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